

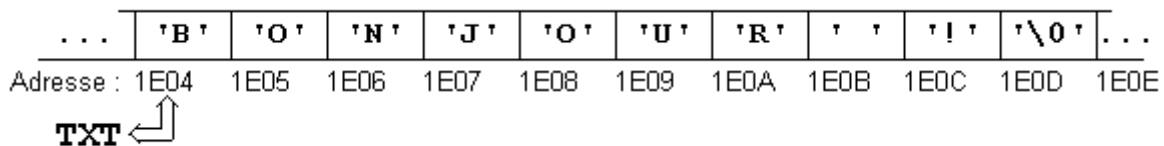
Practical Work No. 13

Character Strings

I. Declaration

Syntax:

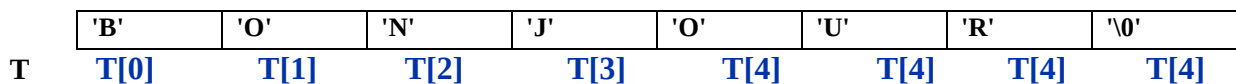
```
char TXT[10] = "BONJOUR !";
```



Accessing an element of a character string can be done in the same way as accessing an element of an array.

Exemple :

```
char T[8] = "BONJOUR";
```



II. Functions from <stdio.h>

The C language provides several ways to read and write strings:

- Using the format specifier `%s` in `printf` and `scanf` functions.
- Using specific functions for reading (`gets`) or displaying (`puts`) a string.

a. The `puts` Function

Syntax: `puts(<string>);`

The `puts` function writes the string specified by `<string>` and automatically adds a newline character at the end.

Example: `puts(txt); /* Equivalent to printf("%s\n", txt); */`

b. The `gets` Function

Syntax: `gets(<string>);`

The `gets` function reads an entire line of characters and copies it to the memory address specified by `<string>`.

III. Functions from `<ctype.h>`

The functions in `<ctype.h>` are used for classifying and converting characters. These functions are independent of the machine's character encoding, enhancing program portability. In the following, `<c>` represents an `int` value that can be interpreted as a character.

The classification functions return a nonzero `int` if the respective condition is met; otherwise, they return 0.

Function	Result
<code>isupper(<c>)</code>	if <code><c></code> is an uppercase letter ('A'...'Z')
<code>islower(<c>)</code>	if <code><c></code> is a lowercase letter ('a'...'z')
<code>isdigit(<c>)</code>	if <code><c></code> is a decimal digit ('0'...'9')
<code>isspace(<c>)</code>	if <code><c></code> is a whitespace character (' ', '\t', '\n', '\r', '\f')

Les fonctions de **conversion** suivantes fournissent une valeur du type **int** qui peut être représentée comme caractère; la valeur originale de `<c>` reste inchangée:

<code>tolower(<c>)</code>	returns <code><c></code> converted to lowercase if <code><c></code> is an uppercase letter
<code>toupper(<c>)</code>	returns <code><c></code> converted to uppercase if <code><c></code> is a lowercase letter

IV. Functions from `<string.h>`

The `<string.h>` library provides a variety of useful functions for string manipulation. Here is a brief description of the most commonly used functions. In the following table:

- `<n>` represents a number of type `int`.
- The symbols `<s>` and `<t>` can be replaced by:
 - a constant string,
 - the name of a variable declared as a `char` array.
- `<c>` represents a character of type `char`.

Function	Result
<code>strlen(<s>)</code>	provides the length of the string excluding the final <code>'\0'</code>
<code>strcat(<s>, <t>)</code>	appends <code><t></code> to the end of <code><s></code>
<code>strncat(<s>, <t>, <n>)</code>	appends at most <code><n></code> characters of <code><t></code> to the end of <code><s></code>
<code>strcmp(<s>, <t>)</code>	compares <code><s></code> and <code><t></code> lexicographically and provides a result: • negative if <code><s></code> precedes <code><t></code> • zero if <code><s></code> is equal to <code><t></code> • positive if <code><s></code> follows <code><t></code>

<code>strcpy(<s>, <t>)</code>	copies <t> to <s>
<code>strncpy(<s>, <t>, <n>)</code>	copies at most <n> characters from <t> to <s>
<code>strchr(<s>, <c>)</code>	searches in the string <s> for the first position where the character <c> appears
<code>strrchr(<s>, <c>)</code>	searches in the string <s> (starting from the end of <s>) for the last position where the character <c> appears
<code>strstr(<s>, <t>)</code>	searches in the string <s> for the first complete occurrence of the substring <t>

V. Functions from <stdlib.h>

The <stdlib.h> library contains function declarations for converting numbers to strings and vice versa.

String --> Number

The three functions defined below are part of the ANSI-C standard and are portable. The symbol <s> can be replaced by:

- a constant string,
- the name of a variable declared as a `char` array.

<code>atoi(<s>)</code>	returns the numeric value represented by <s> as an <code>int</code>
<code>atol(<s>)</code>	returns the numeric value represented by <s> as a <code>long</code>
<code>atof(<s>)</code>	returns the numeric value represented by <s> as a <code>double</code>

General rules for conversion:

- Leading spaces in a string are ignored.
- There is no domain checking for the target.
- The conversion stops at the first non-convertible character.
- For a non-convertible string, the functions return zero.

VI. Application Exercises

Exercise 1:

Write a program that asks for the user's first name and last name and then displays the total length of the name excluding spaces.

Exercise 2:

Write a program that reads two strings CH1 and CH2, compares them lexicographically, and displays the result:

Example:

Enter the first string: ABC
Enter the second string: abc
"ABC" precedes "abc"

Exercise 3:

Write a program that reads two strings CH1 and CH2, and copies the first half of CH1 and the first half of CH2 into a third string CH3. Display the result.

- a) Use the special functions from `<string>`.
- b) Use only the functions `gets` and `puts`.

Exercise 4:

Write a program that reads a French regular verb ending in 'er' from the keyboard and displays its conjugation in the present tense of the indicative. Check if it is indeed a verb ending in 'er' before conjugating

Example: Verb: *fêter*

je fête
tu fêtes
il fête
nous fêtons
vous fêtez
ils fêtent

Exercise 5:

Write a recursive function `Palind` that checks if a string is a palindrome or not. A necessary condition is that the extreme characters are identical, and the substring excluding the extreme characters is also a palindrome.

Exercise 6:

Write a program that reads 10 words and stores them in an array of strings. Sort the 10 words lexicographically using the functions `strcmp` and `strcpy`. Display the sorted array. Use the selection sort method.

Exercise 7:

Write a program that reads 5 words separated by spaces and then displays them in a line, but in reverse order. The words are stored in an array of strings.

Example:

Here a small sentence !
! sentence small a Here